# GIRI SANKAR L

Mechatronics Engineer | Hardware Enthusiast | PCB Design Engineer | Robotics
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## Professional Summary

B.E. Mechatronics Engineering student (2024–2028) passionate about PCB design, hardware development, and embedded systems. Currently developing practical skills in custom PCB design, embedded hardware, and robotics through academic coursework and personal engineering projects. As a hardware enthusiast, I am seeking an ISRO internship to contribute to aerospace and embedded hardware technologies while gaining real-world engineering experience and further strengthening my technical skills

## Technical Skills

**Programming:** C, C++, Python, Embedded C (learning), Arduino Programming

**PCB Design:** EasyEDA Pro, Ki-CAD, Altium Designer, Schematic Design, PCB Layouts, 4-Layer PCB Design

**Embedded Systems:** Arduino Uno, RP2040, ESP32, Sensors & Actuators, Motor Drivers, Servo Motors, STM32 based PCB’s and codings (learning)

**Hardware Design:** Component Selection, Circuit Debugging, Hardware Prototyping, Electronic Circuit Design, Component Selection, Component Selection, Design for Manufacturing (DFM)

**Tools:** Git, GitHub (Learning), VS Code

**Areas of Interest:** PCB Design, Hardware Design, Robotics, Electronics, Industrial Automation, Embedded Systems (learning)

## Professional Experience

### Team Leader

*ABU Robocon 2026 | 2026*

- Participated as a team leader in the International-level ABU Robocon 2026 and the team shortlisted for National Selection.
- Contributed to hardware development, embedded programming, and robotic system integration.

*Robotics innovation Hackathon 2026 | 2026*

- Idea submitted for the round 1 and waiting for the result conducted by IEEE Robotic Society.

## Projects

**Custom 4-Layer RP2040 Development Board**

- Designed a compact 4-layer RP2040 development board inspired by the Seeed Studio XIAO RP2040.
- Built full schematic, BOM, and PCB layout in EasyEDA Pro; verified with ERC and DRC checks
- Next phase: PCB fabrication and hardware bring-up/testing.

**Universal Embedded Mini Controller (UMC Board)***

- Designed a compact 40 mm × 40 mm embedded controller board for general-purpose embedded applications with modular, expandable architecture.
- Planned as a compact embedded motherboard suitable for multiple robotics and IoT applications.

**Autonomous Seed Sowing Robot**

- Built a working prototype for automated seed sowing; planned next phase includes SLAM-based mapping and autonomous navigation

**AI-Based Warehouse Automation**

- Authored and submitted an automation concept proposal to RIH, an event conducted by the IEEE Robotics Society

**Obstacle Avoidance Robot & Flame Detection Robot**

- Built sensor-driven robots as part of early hands-on robotics and embedded systems learning

**Arduino Light Control PCB / PCB Design Practice**

- Designed and practiced schematic-to-PCB workflows, building foundational PCB layout skills

## Education

**B.E. Mechatronics Engineering**

*Hindusthan College of Engineering and Technology, Tamil Nadu | 2024 – 2028*

**CGPA: 8.4 / 10**

## Achievements & Certifications

- NPTEL Elite Certification — "Introduction to Industry 4.0 and Industrial Internet of Things", **IIT Kharagpur (Jan–Apr 2026)**
- Team Leader, National-level ABU Robocon 2026 competition.